Hamiltonian Constraint for Modified Gravity

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Hamiltonian Constraint $\mathcal{H} = 0$ correspond to Friedmann Equation to make sure the temporal diffeomorphism invariant valid. In our modified gravity case, we found in ghost free theory, the modified friedmann equation itself has no dependence on second or higher derivative varible. Therefore, this can be served as a constrain to make temporal diffeomorphism invariant preserved in this case.

I. MODIFIED GRAVITY AND FRIEDMANN CONSTRAIN

We clarify the status of the lapse equation in the FLRW minisuperspace reduction. Consider a generally covariant action reduced on the spatially flat FLRW ansatz

$$ds^2 = -N(t)^2 dt^2 + a(t)^2 d\vec{x}^2, \tag{1}$$

possibly coupled to a homogeneous scalar field $\phi(t)$. The reduced action takes the form

$$S = \int dt\, L\big(N, \dot{N}, \ddot{N}, \ldots; a, \dot{a}, \ddot{a}, \ldots; \phi, \dot{\phi}, \ddot{\phi}, \ldots\big). \tag{2}$$

We define the Euler–Lagrange expressions by

$$\delta S = \int dt\, \left(E_N\, \delta N + E_a\, \delta a + E_\phi\, \delta\phi\right), \tag{3}$$

where, for a variable q,

$$E_q = \sum_{k\geq 0} (-1)^k \frac{d^k}{dt^k}\left(\frac{\partial L}{\partial q^{(k)}}\right). \tag{4}$$

a. Residual time-diffeomorphism invariance. Because the parent four-dimensional action is generally covariant, the FLRW-reduced action is invariant, up to a boundary term, under the residual time reparametrization

$$t \mapsto t - \xi(t), \tag{5}$$

under which the minisuperspace fields transform as

$$\delta_\xi a = \xi \dot{a}, \qquad \delta_\xi \phi = \xi \dot{\phi}, \qquad \delta_\xi N = (N\xi)^{\cdot} = \dot{N}\,\xi + N\dot{\xi}. \tag{6}$$

Here we use the active-sign convention for the infinitesimal transformation. With the opposite convention, all variations acquire an overall minus sign and the final Noether identity is unchanged up to an irrelevant overall sign.

The invariance of the action implies

$$0 = \delta_\xi S = \int dt\, \Big[E_N (N\xi)^{\cdot} + E_a \dot{a}\,\xi + E_\phi \dot{\phi}\,\xi\Big]. \tag{7}$$

Expanding the first term gives

$$E_N (N\xi)^{\cdot} = E_N \dot{N}\,\xi + N E_N \dot{\xi}. \tag{8}$$

Integrating the second term by parts,

$$-\int dt\, N E_N \dot{\xi} = \int dt\, \frac{d}{dt}(N E_N)\xi = \tag{9}$$

$$\int dt\, \Big(\dot{N} E_N + N \dot{E}_N\Big)\,\xi, \tag{10}$$

and assuming that $\xi(t)$ vanishes at the time boundaries, we obtain

$$0 = \int dt\, \Big[-N\dot{E}_N + \dot{a}E_a + \dot{\phi}E_\phi\Big]\,\xi. \tag{11}$$

Since $\xi(t)$ is arbitrary, the integrand must vanish identically:

$$\boxed{N\dot{E}_N - \dot{a}E_a - \dot{\phi}E_\phi = 0.} \tag{12}$$

This is the minisuperspace Noether identity associated with residual time-diffeomorphism invariance. It is the FLRW version of the contracted Bianchi identity. Importantly, Eq. (12) is an off-shell identity; it holds without imposing any equation of motion.

b. Consequence for the lapse equation. We now show that, in the ghost-free FLRW sector, the lapse equation is a constraint rather than an independent dynamical equation.

Let us work in cosmic-time gauge after variation,

$$N = 1, \qquad \dot{N} = \ddot{N} = \cdots = 0. \tag{13}$$

Then Eq. (12) becomes

$$\dot{E}_N\big|_{N=1} = \Big(\dot{a}E_a + \dot{\phi}E_\phi\Big)\big|_{N=1}. \tag{14}$$

Assume that, after imposing the ghost-free coefficient relations, the gauge-fixed dynamical equations $E_a|_{N=1} = 0$ and $E_\phi|_{N=1} = 0$ are at most second order in time derivatives. That is,

$$E_a\big|_{N=1} = E_a(a, \dot{a}, \ddot{a}, \phi, \dot{\phi}, \ddot{\phi}), \qquad E_\phi\big|_{N=1} = E_\phi(a, \dot{a}, \ddot{a}, \phi, \dot{\phi}, \ddot{\phi}), \tag{15}$$

with no $a^{(3)}$, $\phi^{(3)}$, or higher time derivatives. Then the right-hand side of Eq. (14) contains at most second derivatives.

Suppose, for contradiction, that $E_N|_{N=1}$ contains an independent acceleration. Locally in jet space, we may write its acceleration-dependent part as

$$E_N\big|_{N=1} = F_0(a, \dot{a}, \phi, \dot{\phi}) + F_a(a, \dot{a}, \phi, \dot{\phi})\,\ddot{a} + F_\phi(a, \dot{a}, \phi, \dot{\phi})\,\ddot{\phi} + \cdots, \tag{16}$$

where the ellipsis denotes possible terms nonlinear in $\ddot{a}$ and $\ddot{\phi}$. More generally, $E_N|_{N=1}$ may be viewed as a function

$$E_N\big|_{N=1} = \mathcal{C}(a, \dot{a}, \ddot{a}, \phi, \dot{\phi}, \ddot{\phi}). \tag{17}$$

Taking a time derivative gives

$$\dot{E}_N\big|_{N=1} = \frac{\partial \mathcal{C}}{\partial \ddot{a}}\, a^{(3)} + \frac{\partial \mathcal{C}}{\partial \ddot{\phi}}\, \phi^{(3)} + \text{terms containing at most second derivatives.} \tag{18}$$

However, Eq. (14) states that $\dot{E}_N|_{N=1}$ is equal to $(\dot{a}E_a + \dot{\phi}E_\phi)|_{N=1}$, which contains no independent third derivatives under the ghost-free assumption. Since the identity is off shell and must hold for arbitrary field profiles, the independent jet coordinates $a^{(3)}$ and $\phi^{(3)}$ cannot appear. Therefore

$$\frac{\partial \mathcal{C}}{\partial \ddot{a}} = 0, \qquad \frac{\partial \mathcal{C}}{\partial \ddot{\phi}} = 0. \tag{19}$$

Hence $E_N|_{N=1}$ is independent of the accelerations:

$$\boxed{E_N\big|_{N=1} = \mathcal{C}_N(a, \dot{a}, \phi, \dot{\phi}).} \tag{20}$$

Thus the lapse equation

$$E_N\big|_{N=1} = \mathcal{C}_N(a, \dot{a}, \phi, \dot{\phi}) = 0 \tag{21}$$

is a first-order constraint but not a dynamical equation on the FLRW initial data, rather than an independent second-order evolution equation. It is therefore the modified Friedmann constraint of the candidate theory. And the diffeomorphism invariant still preserved.

c. Constraint preservation. The same Noether identity also guarantees preservation of the constraint. On solutions of the dynamical equations,

$$E_a = 0, \qquad E_\phi = 0, \tag{22}$$

which is the on-shell condition imposed to Eq. (14) giving

$$\dot{E}_N\big|_{N=1} = 0. \tag{23}$$

Therefore, if the initial data satisfy the lapse equation at some initial time t_0,

$$E_N(t_0)\big|_{N=1} = 0, \tag{24}$$

then the constraint remains satisfied along the entire background evolution:

$$E_N(t)\big|_{N=1} = 0. \tag{25}$$

This justifies using $E_N = 0$ as the modified Friedmann constraint for setting initial conditions, while using the independent second-order equations $E_a = 0$ and $E_\phi = 0$ to evolve the background.

d. Relation to the Hamiltonian constraint. The above result should be interpreted carefully. In ordinary GR, after removing the boundary term from the Einstein–Hilbert action, the FLRW minisuperspace Lagrangian can be written as

$$L_{\rm EH} = -3M_{\rm Pl}^2 V_0\, \frac{a\dot{a}^2}{N}. \tag{26}$$

It contains no $\dot{N}$, so the momentum conjugate to the lapse vanishes,

$$p_N = \frac{\partial L_{\rm EH}}{\partial \dot{N}} = 0. \tag{27}$$

This is a primary constraint. Preservation of $p_N = 0$ yields the Hamiltonian constraint

$$\mathcal{H}_{\rm GR} = 0. \tag{28}$$

In FLRW, this Hamiltonian constraint is precisely the usual Friedmann constraint.

But for a general modified-gravity action, the raw FLRW-reduced Lagrangian may contain $\dot{N}$ and higher derivatives before integrations by parts. The presence of $\dot{N}$ in such a raw higher-derivative representation does not by itself prove that the lapse is a propagating variable, because $\dot{N}$ may be removable together with higher-derivative terms by boundary terms or by passing to an equivalent first-order formulation. Conversely, proving the identity

$$N\dot{E}_N - \dot{a}E_a - \dot{\phi}E_\phi = 0 \tag{29}$$

does not by itself constitute a full ADM/Dirac Hamiltonian analysis of the four-dimensional covariant theory.

What the minisuperspace Noether identity proves is the following:

In the FLRW ghost-free sector, the lapse equation $E_N = 0$

is a preserved first-order modified Friedmann constraint.

Equivalently, $E_N = 0$ is the FLRW Lagrangian counterpart of the modified Hamiltonian constraint on the homogeneous background.

A stronger statement would require a full Hamiltonian analysis of the covariant theory, including the lapse and shift, the primary and secondary constraints, and the closure of the full constraint algebra:

$$\{\mathcal{H}(x), \mathcal{H}(y)\}, \qquad \{\mathcal{H}_i(x), \mathcal{H}(y)\}, \qquad \{\mathcal{H}_i(x), \mathcal{H}_j(y)\}. \tag{30}$$

Therefore, the present argument establishes background-level consistency of the modified Friedmann equation, but it should not be overstated as a proof of the complete nonperturbative Hamiltonian constraint algebra of the full modified-gravity theory.